

RTL Design Naming Rule

This document defines mandatory naming rules for RTL design. The rules apply to modules, instances, ports, internal signals, constants, and integration interfaces. Exceptions shall be explicitly reviewed and agreed before implementation.

V1.0

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Revision

Version	Date	Author	Description
1.0	15/09/2026	Tam Nguyen	Initial version

1. GENERAL DESIGN PRINCIPLES

1.1. Identifier Format

All RTL identifiers shall:

- Use lowercase English letters.
- Use digits where required.
- Use underscore _ as the only word separator.
- Not use CamelCase, PascalCase, or mixed-case naming.
- Use meaningful and consistent terminology throughout the project.

Examples

Valid	Invalid
m_qnsc_timer	M_QNSC_TIMER
i_bus_apb_paddr	iApbAddr
o_int_timer_0	o_irq_timer0
i_rst_n_sys	rstn_sys
u_timer_0	uTimer0

1.2. Port Direction Prefix

All module ports shall use the following prefixes:

Prefix	Meaning	Example
i_	Input	i_clk_sys
o_	Output	o_int_timer_0
io_	Bidirectional	io_i2c_sda

The direction prefix shall always be placed at the beginning of the port name.

1.3. Active-Low Naming

Active-low signals shall contain `_n` immediately after the signal meaning.

Examples:

```
i_rst_n_sys  
i_rst_n_cpu  
o_rst_n_peri
```

`_n` shall not be replaced by inconsistent forms such as:

```
rstn  
resetn  
reset_b  
rst_bar
```

1.4. Numeric Instance and Channel Naming

Numeric instances or channels shall use an underscore before the index.

Valid

```
timer_0  
timer_1  
uart_0  
uart_1  
dma_0  
dma_1
```

Invalid

```
timer0  
timer1  
uart0  
dma0
```

1.5. Standard Vocabulary

Project-wide standard vocabulary shall be used consistently.

Function	Mandatory terminology
Interrupt	int
Power	pwr
Design-for-Test	dft
Analog	ana
Multiplexer	mux
Clock	clk
Reset	rst
Configuration	cfg
Debug	dbg
Test	test
Boot	boot
Memory	mem
Peripheral	peri
UART	uart
SPI	spi
I2C	i2c
GPIO	gpio
DMA	dma

Standard protocol vocabulary shall be preserved, including:

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paddr
pwrdata
prdata
pready
psel
penable
pwrite

awaddr
awvalid
awready
wdata
wstrb
wvalid
wready
bresp
bvalid
bready
araddr
arvalid
arready
rdata
rresp
rvalid
rready

2. MODULE, INSTANCE, AND INTERNAL RTL NAMING

2.1. Module Naming

QNSC functional modules shall use:

```
m_qnsc_<function>
```

Examples:

```
m_qnsc_timer  
m_qnsc_dma  
m_qnsc_gpio  
m_qnsc_uart
```

QNSC wrapper modules shall use:

```
m_qnsc_wrap_<ip_module>
```

Examples:

```
m_qnsc_wrap_ibex  
m_qnsc_wrap_apb_uart  
m_qnsc_wrap_timer
```

Generic modules shall use:

```
qnsc_<function>
```

Examples:

```
qnsc_fifo_sync  
qnsc_arbiter  
qnsc_clock_gate  
qnsc_reset_sync
```

2.2. Instance Naming

Module instances shall use:

```
u_<function>_<index>
```

Examples:

```
u_timer_0
```

```
u_timer_1
u_uart_0
u_dma_0
u_fifo_0
```

For a single non-indexed instance:

```
u_timer
u_cdc
u_reset_sync
```

2.3. Internal Signal Naming

Internal registered signals shall use:

```
r_<function>
```

Internal combinational signals shall use:

```
w_<function>
```

Examples:

```
r_timer_count
r_timer_enable
r_state
```

```
w_timer_done
w_timer_match
w_next_state
```

Example

```
reg [31:0] r_timer_count;
wire      w_timer_done;
```

2.4. Parameter, Constant, and State Naming

Parameters shall use:

```
P_<FUNCTION>
```

Examples:

```
P_DATA_WIDTH
P_ADDR_WIDTH
P_DEPTH
```

Local constants shall use:

C_<FUNCTION>

Examples:

C_IDLE
C_BUSY
C_ERROR

FSM states shall use:

S_<STATE>

Examples:

S_IDLE
S_WAIT
S_WRITE
S_READ

Example:

```
localparam P_DATA_WIDTH = 32;  
  
localparam C_IDLE  = 2'd0;  
localparam C_BUSY  = 2'd1;
```

2.5. Memory Naming

Memory arrays shall use:

mem_<function>

Examples:

mem_data
mem_tag
mem_fifo
mem_instruction

3. PORT AND INTERFACE NAMING

3.1. Clock

Clock ports shall use:

```
i_clk_<domain>  
o_clk_<domain>
```

Examples:

```
i_clk_sys  
i_clk_cpu  
i_clk_peri  
o_clk_peri
```

3.2. Reset

Reset ports shall use:

```
i_rst_n_<domain>  
o_rst_n_<domain>
```

Examples:

```
i_rst_n_sys  
i_rst_n_cpu  
i_rst_n_peri  
o_rst_n_peri
```

3.3. APB

APB ports shall use:

```
i_bus_apb_<signal>  
o_bus_apb_<signal>
```

Examples:

```
i_bus_apb_paddr  
i_bus_apb_psel  
i_bus_apb_penable  
i_bus_apb_pwrite  
i_bus_apb_pwdata
```

```
o_bus_apb_prdata
o_bus_apb_pready
o_bus_apb_pslverr
```

3.4. AXI

AXI naming shall be normalized by individual channel and signal.

The format shall be:

```
i_bus_axi_<channel>_<signal>
o_bus_axi_<channel>_<signal>
```

3.4.1. AW Channel

```
i_bus_axi_aw_addr
i_bus_axi_aw_valid
o_bus_axi_aw_ready
```

3.4.2. W Channel

```
i_bus_axi_w_data
i_bus_axi_w_strb
i_bus_axi_w_valid
o_bus_axi_w_ready
```

3.4.3. B Channel

```
o_bus_axi_b_resp
o_bus_axi_b_valid
i_bus_axi_b_ready
```

3.4.4. AR Channel

```
i_bus_axi_ar_addr
i_bus_axi_ar_valid
o_bus_axi_ar_ready
```

3.4.5. R Channel

```
o_bus_axi_r_data
o_bus_axi_r_resp
o_bus_axi_r_valid
i_bus_axi_r_ready
```

3.5. CSR

CSR ports shall use:

```
i_csr_<function>
o_csr_<function>
```

Examples:

```
i_csr_timer_enable  
i_csr_timer_prescaler  
  
o_csr_timer_status  
o_csr_timer_count
```

3.6. Interrupt

Interrupt ports shall use:

```
i_int_<source>  
o_int_<source>
```

Examples:

```
o_int_timer_0  
o_int_uart_0  
o_int_dma_0  
  
i_int_external  
i_int_software  
i_int_timer
```

3.7. GPIO

GPIO ports shall use:

```
i_gpio_<function>  
o_gpio_<function>  
io_gpio_<function>
```

Examples:

```
i_gpio_uart_rx  
o_gpio_spi_sclk  
o_gpio_i2c_sda
```

3.8. Pad

Pad ports shall use:

```
i_pad_<function>  
o_pad_<function>  
io_pad_<function>
```

Examples:

```
i_pad_usi_0  
o_pad_can_sel  
io_pad_i2c_scl
```

3.9. Memory

Memory ports shall use:

```
i_mem_<function>  
o_mem_<function>
```

Examples:

```
o_mem_addr  
o_mem_wdata  
o_mem_we  
o_mem_be  
i_mem_rdata
```

3.10. DMA

DMA ports shall use:

```
i_dma_<function>  
o_dma_<function>
```

Examples:

```
o_dma_req  
i_dma_gnt  
i_dma_rvalid  
o_dma_addr  
o_dma_wdata  
i_dma_rdata
```

3.11. JTAG

JTAG ports shall use:

```
i_jtag_<function>  
o_jtag_<function>
```

Examples:

```
i_jtag_tck
```

```
i_jtag_tms  
i_jtag_tdi  
o_jtag_tdo
```

3.12. Debug

Debug ports shall use:

```
i_dbg_<function>  
o_dbg_<function>
```

Examples:

```
i_dbg_req  
i_dbg_enable  
o_dbg_ack  
o_dbg_halted
```

3.13. Boot

Boot ports shall use:

```
i_boot_<function>  
o_boot_<function>
```

Examples:

```
i_boot_addr  
i_boot_sel  
o_boot_done
```

3.14. Power

Power-related ports shall use:

```
i_pwr_<function>  
o_pwr_<function>
```

Examples:

```
i_pwr_sleep  
i_pwr_isolation  
o_pwr_good
```

3.15. DFT

DFT ports shall use:

```
i_dft_<function>  
o_dft_<function>
```

Examples:

```
i_dft_scan_en  
i_dft_bypass  
o_dft_scan_out
```

3.16. Analog

Analog ports shall use:

```
i_ana_<function>  
o_ana_<function>  
io_ana_<function>
```

Examples:

```
i_ana_comp_out  
o_ana_trim  
io_ana_test
```

3.17. Test

Test-related ports shall use:

```
i_test_<function>  
o_test_<function>
```

Examples:

```
i_test_en  
i_test_mode  
o_test_status
```

4. COMPLETE EXAMPLES AND ENFORCEMENT

4.1. Complete QNSC Wrapper Example

```
module m_qnsc_wrap_timer (
    input  wire          i_clk_sys,
    input  wire          i_rst_n_sys,

    input  wire [15:0] i_bus_apb_paddr,
    input  wire          i_bus_apb_psel,
    input  wire          i_bus_apb_penable,
    input  wire          i_bus_apb_pwrite,
    input  wire [31:0] i_bus_apb_pwdata,

    output wire [31:0] o_bus_apb_prdata,
    output wire          o_bus_apb_pready,
    output wire          o_bus_apb_pslverr,

    output wire          o_int_timer_0
);
```

Internal RTL:

```
reg [31:0] r_timer_count;
reg          r_timer_enable;

wire          w_timer_done;
wire          w_timer_match;

localparam P_DATA_WIDTH = 32;

localparam C_IDLE = 2'd0;
localparam C_BUSY = 2'd1;

reg [1:0] r_state;
```

Instance:

```
timer u_timer_0 (
    ...
);
```

4.2. Complete Naming Example

A timer module integrated through APB shall follow:

```
Module
    m_qnsc_wrap_timer

Clock
    i_clk_sys

Reset
```

```
i_rst_n_sys
```

APB

```
i_bus_apb_paddr
i_bus_apb_psel
i_bus_apb_penable
i_bus_apb_pwrite
i_bus_apb_pwdata
o_bus_apb_prdata
o_bus_apb_pready
o_bus_apb_pslverr
```

Interrupt

```
o_int_timer_0
```

Internal registered signal

```
r_timer_count
```

Internal combinational signal

```
w_timer_done
```

Parameter

```
P_DATA_WIDTH
```

Constant

```
C_IDLE
```

Instance

```
u_timer_0
```

4.3. Enforcement Checklist

Before RTL is committed, the following rules shall be satisfied:

Check	Mandatory requirement
Identifier case	All identifiers shall be lowercase
Word separator	Words shall use _
Port direction	Ports shall use i_, o_, or io_
QNSC module	Shall use m_qnsc_<function>
QNSC wrapper	Shall use m_qnsc_wrap_<ip_module>
Instance	Shall use u_<function>_<index> where indexed

Check	Mandatory requirement
Registered signal	Shall use <code>r_<function></code>
Combinational signal	Shall use <code>w_<function></code>
Parameter	Shall use <code>P_<FUNCTION></code>
Constant	Shall use <code>C_<FUNCTION></code>
FSM state	Shall use <code>S_<STATE></code>
Numeric instance	Shall use <code>timer_0</code> , not <code>timer0</code>
AXI	Shall be normalized by channel and signal
Pad	Shall use <code>i_pad_</code> , <code>o_pad_</code> , <code>io_pad_</code>
Power	Shall use <code>pwr</code>
DFT	Shall use <code>dft</code>
Analog	Shall use <code>ana</code>
Mux	Shall use <code>mux</code>
Third-party IP	Shall be normalized at wrapper boundary